Lab 4

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Type	Laboratory
Reviewed	☐

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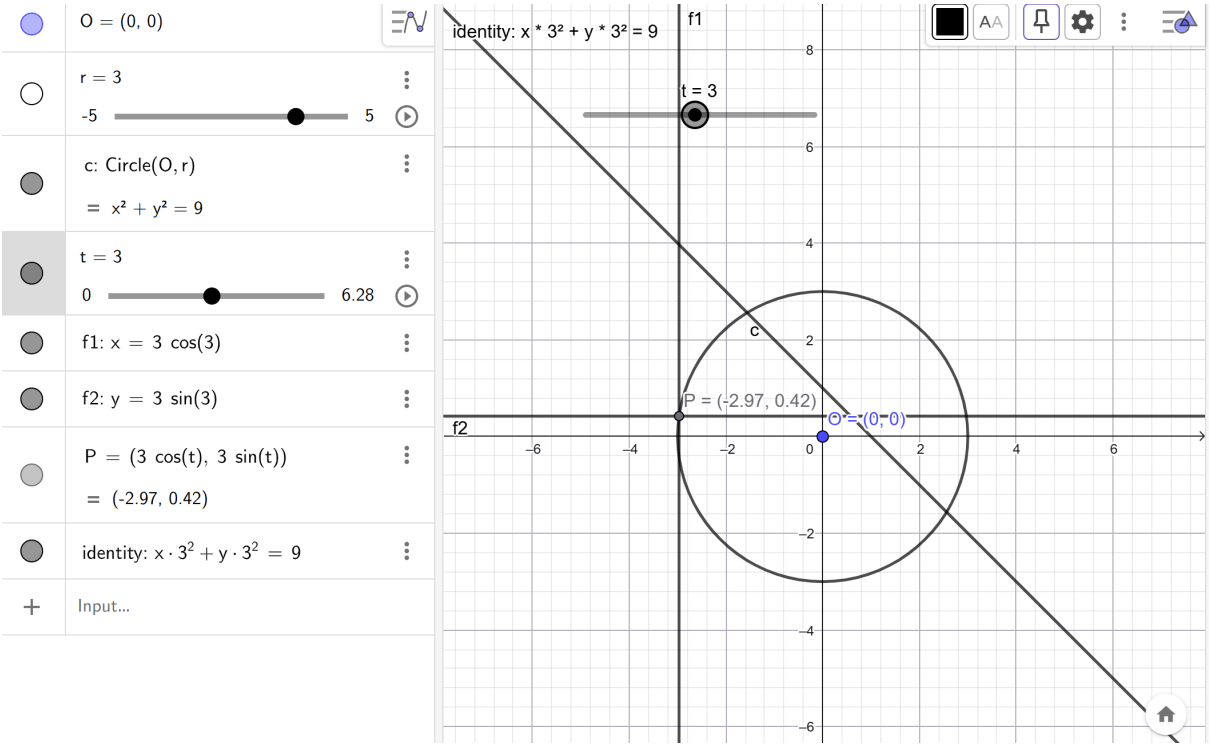
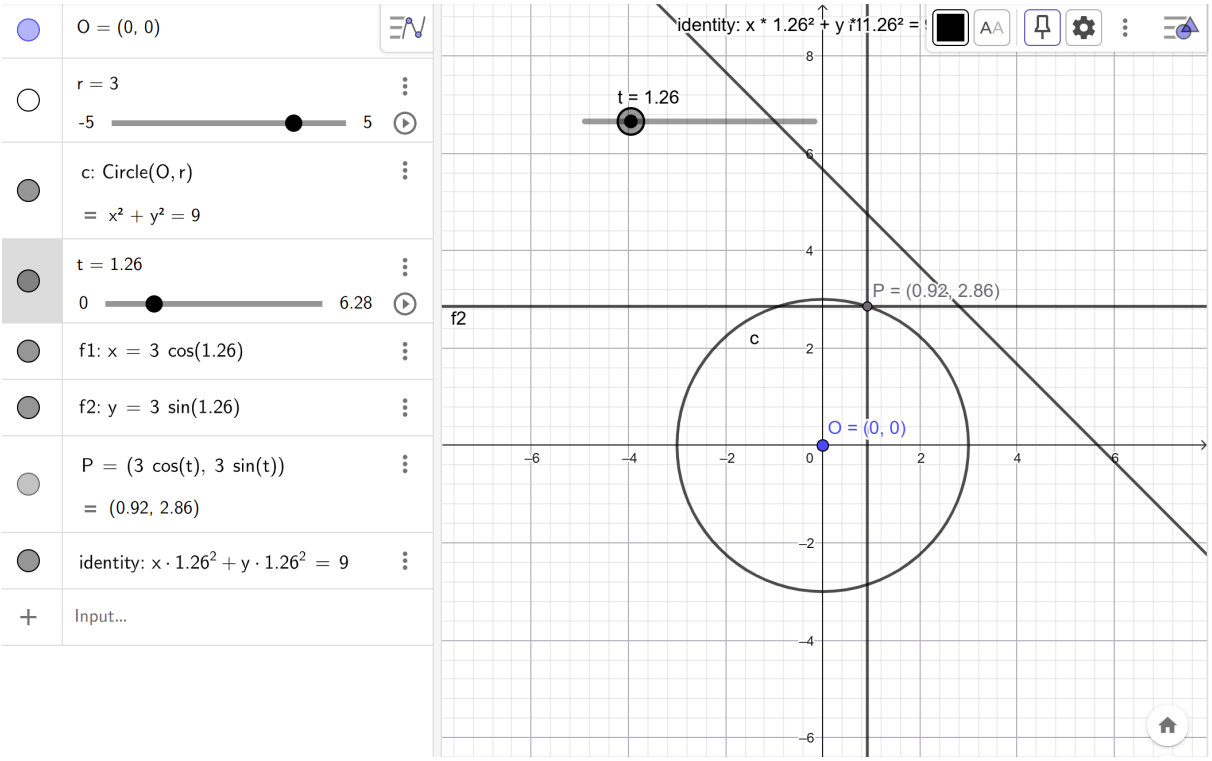
Group	Group 1 (attending Group 3 lab)
Major	Artificial Intelligence

Exercise 1

Construct in GeoGebra the circle with center O(0, 0) and radius 3:
● Use the parametric representation x(t) = 3 cos t, y(t) = 3 sin t, for 0 <= t <= 2π.
● Animate a point P(t) on the circle.
● Add dynamic text showing: the current value of t, the coordinates of P(t), and the identity
x(t)^2 + y(t)^2 = 9. What to submit: Submit the curve, the moving point, the dynamic text, and one sentence explaining why the motion is periodic.





There's dynamic text and we can see point P at intersection f1 f2 is animated.

The motion is periodic because the parametrization uses sine and cosine, which repeat every 2π, so the point returns to its starting position after one full revolution.

https://www.geogebra.org/classic/xjnkevey

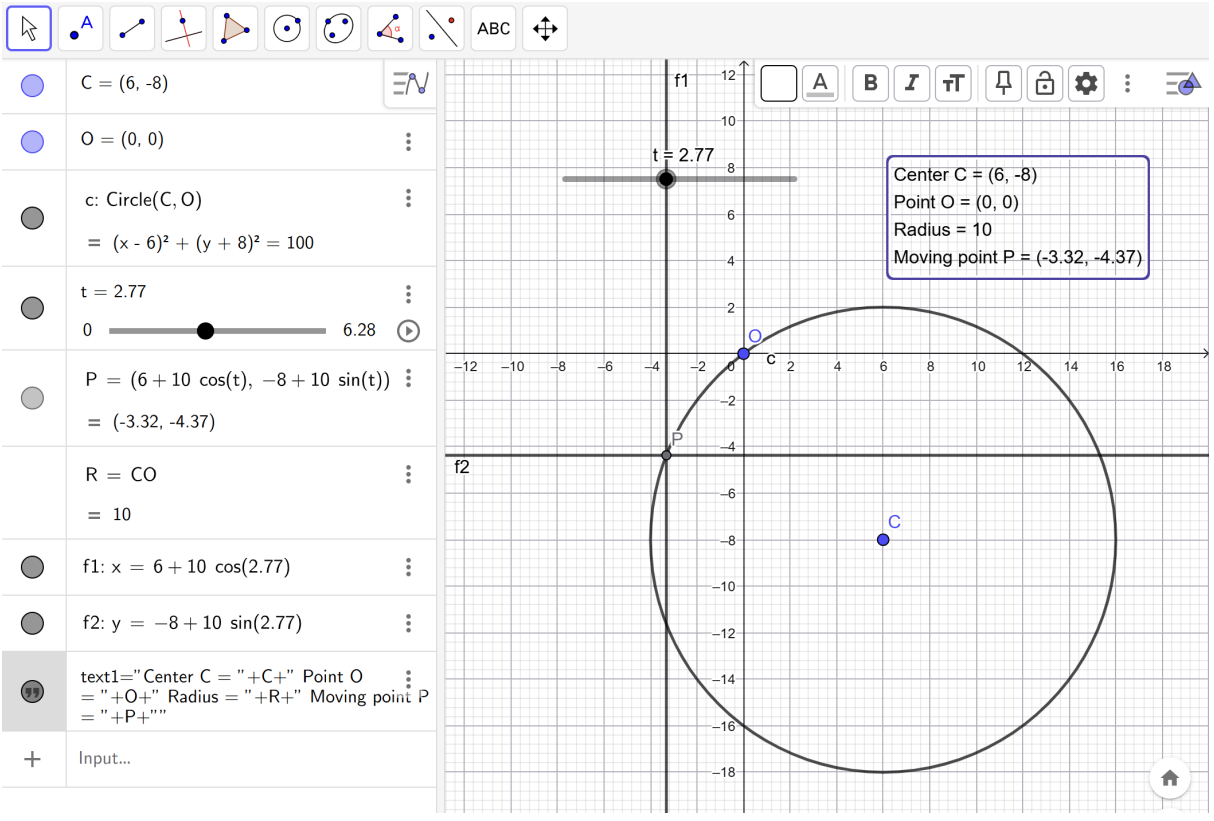
PS: I have added dynamic text as well, check link

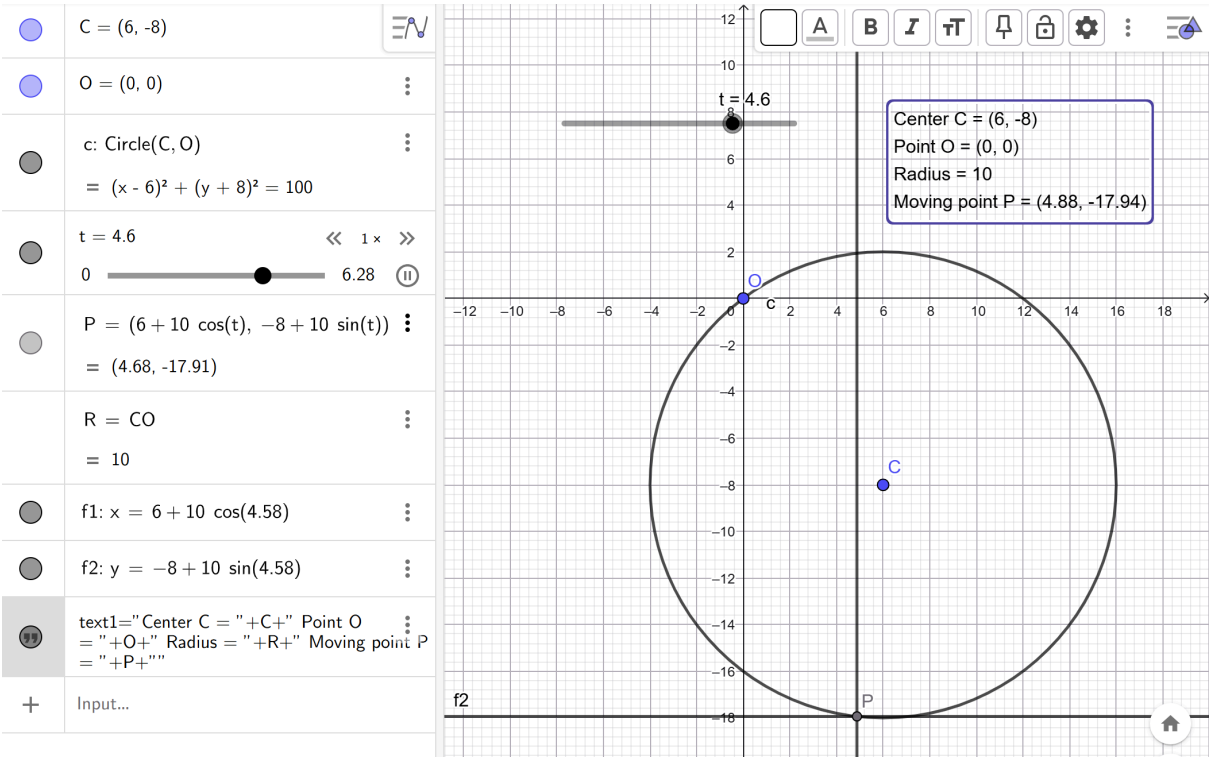
Exercise 2

Construct the circle with center C(6, − 8) and passing through the origin O(0, 0).

● First compute the radius, then use a parametric representation of the form x(t) = 6 + R cos t, y(t) = − 8 + R sin t, for 0 <= t <= 2π.

● Animate a point P(t) on the circle and show dynamically the center, radius, and coordinates of the
moving point.





As you can see the point P is animated and the text is dunamic

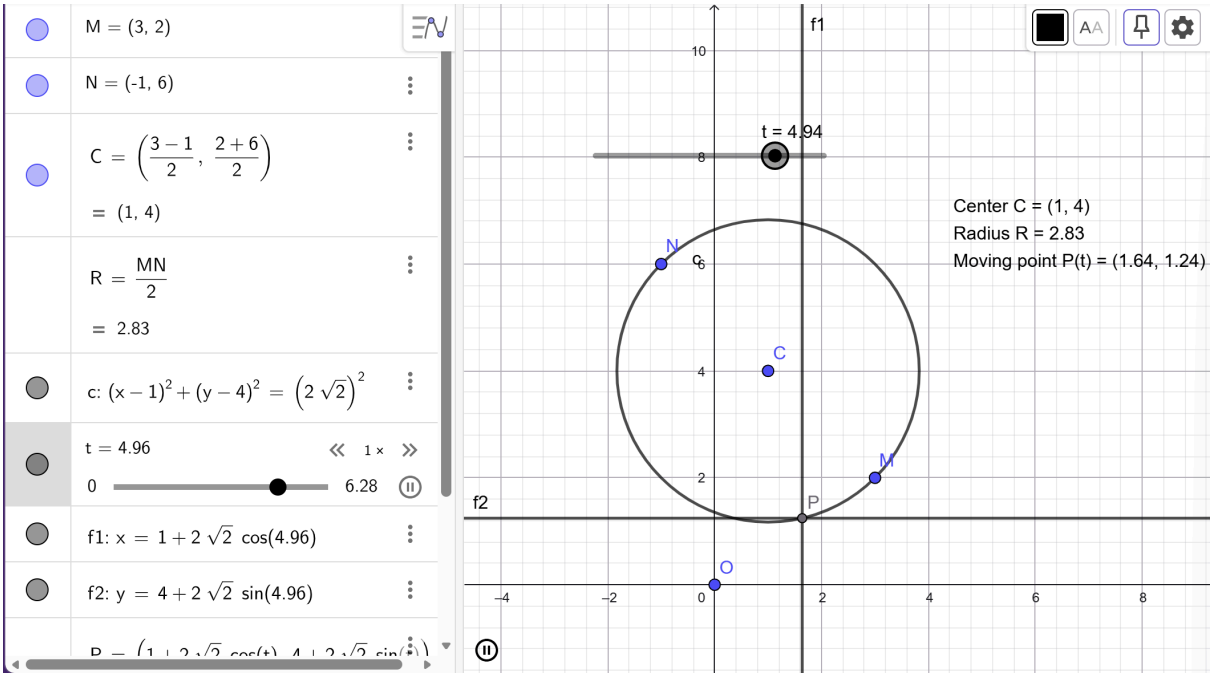
https://www.geogebra.org/classic/tpzsmznd

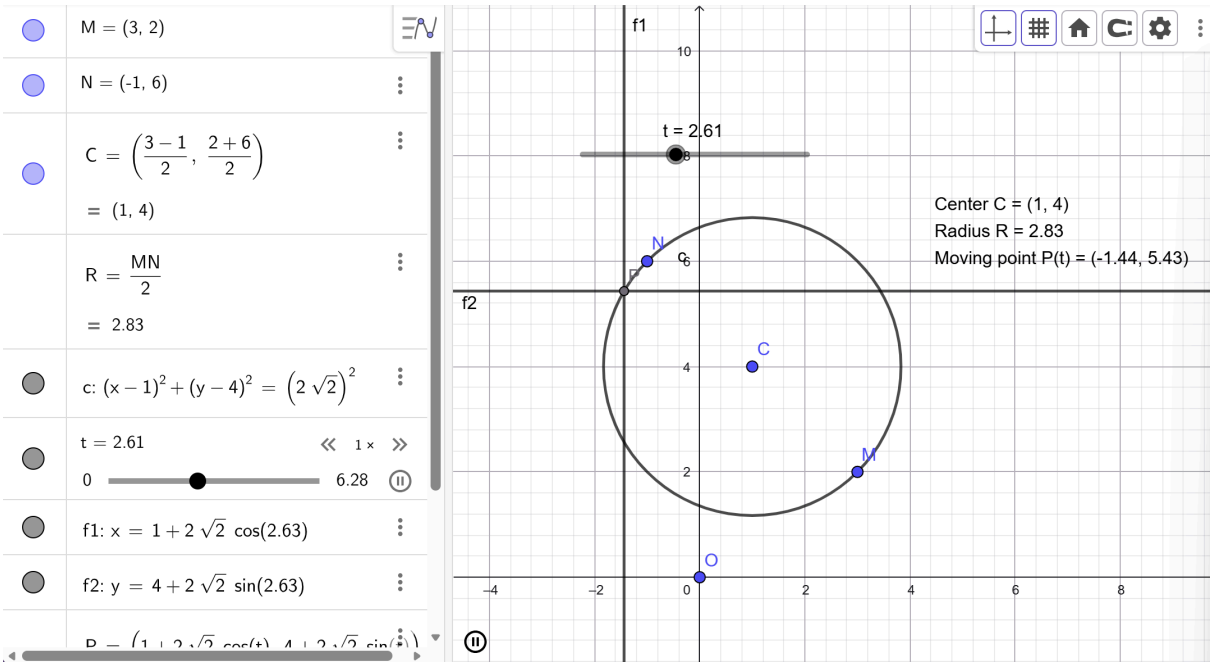
Exercise 3

- Let M(3, 2) and N(− 1, 6) be the endpoints of a diameter.
- Construct the corresponding circle in GeoGebra and write it in parametric form

x(t) = x_C + R cos t, y(t) = y_C + R sin t , for 0 <= t <= 2π

- Animate a point P(t) on the circle.
- Add dynamic text showing the center C, the radius R, and the coordinates of P(t).





You can see point P(t) is animated and moving in accordance with value of t.

https://www.geogebra.org/classic/xqdqqy3y

Exercise 4

Exercise 5

Exercise 6